Midterm Project · Milestone 2: Baseline Linear Regression

A Baseline Linear Model for Daily Bike-Share Demand

[Your name] · [Course] · October 2026 · Analysis notebook: *m2BaselineRegression.jl* (Pluto)

Executive summary

What we did. We used two years (2011–2012) of daily records from Capital Bikeshare in Washington, D.C., to build a statistical model of how the weather and the calendar relate to the number of bikes rented each day. The model accounts for about 88 % of the day-to-day ups and downs in rentals.

What we found. Each comparison below holds the other factors fixed.

- **Temperature is the strongest weather factor.** Demand climbs steadily from freezing temperatures to a peak at about 22 °C (72 °F), where an average day sees roughly 5,700 rentals. It then falls off on hot days: a 30 °C (86 °F) day has about 1,200 fewer rentals than a day at the peak.
- **Rain or snow is costly**, with about 1,700 fewer rentals than a comparable clear day. Misty or cloudy weather costs about 370.
- **Humidity and wind also lower demand**: roughly 240 fewer rentals for every 10 points of humidity, and 520 fewer for every 10 km/h (6 mph) of wind.
- **Holidays draw about 620 fewer rentals than ordinary working days.** Weekends are about the same as working days.
- **Use grew quickly**: 2012 averaged about 1,950 more rentals per day than 2011.

How much to trust it. A typical day's prediction is off by about 690 rentals. The model is least reliable on unusual days, such as two hurricanes, Thanksgiving and Christmas, and on very cold, wet days, when it can even predict fewer than zero rentals.

What comes next. Holidays do not all behave alike. Thanksgiving 2012 drew about 2,900 fewer riders than the model predicted, while the warm Independence Day of 2011 drew about 1,800 more. The next milestone will test whether the holiday effect depends on temperature, and the final project will use models designed for counts.

1. Data and question

The daily file of the UCI *Bike Sharing* dataset (Fanaee-T, 2013; Fanaee-T & Gama, 2013) records Capital Bikeshare rentals in Washington, D.C., on 731 days in 2011–2012. The outcome is total rentals per day (*cnt*: mean 4,510, SD 1,933). The predictors are temperature (°C), relative humidity (%), wind speed (km/h), weather situation (clear; mist or cloudy; light rain or snow), season, year, and a **day type** factor (working day, weekend, holiday) that replaces the overlapping *holiday*, *weekday* and *workingday* flags. Weather variables were converted to physical units (temperature = −8 + 47 × *temp*), and season codes were relabeled because code 1 is winter. One day with an impossible 0 % humidity during rain (March 10, 2011) was removed, leaving **n = 730**. This baseline model is the starting point for Milestone 3, which asks whether the holiday effect is meaningful after adjusting for weather and whether it depends on temperature.

2. Building the model

A simple regression on temperature gives $\hat{y}$ = 2,353.8 + 141.00·Temp (R^2 = 0.394). With $t_{0.995,\,728}$ = 2.583, the 99 % confidence interval for the slope is 141.0 ± 2.583 × 6.48 = (124.3, 157.7) rentals per °C. Adding the other weather and calendar terms raises R^2 to 0.822, but the residuals of that model, plotted against temperature, form a clear arch (Figure 1a): the straight line over-predicts cool and hot days and under-predicts mild ones. Polynomial terms in the centered temperature *T* = temperature − 20 °C fix this. The nested F-test of the linear against the cubic model (H_0: $\beta_{T^2} = \beta_{T^3} = 0$) gives F = 158.0 on 2 and 716 df ($p < 0.001$), while a quartic term adds nothing ($p = 0.318$). The cubic model also has the lowest AIC and BIC (Appendix Table A1).

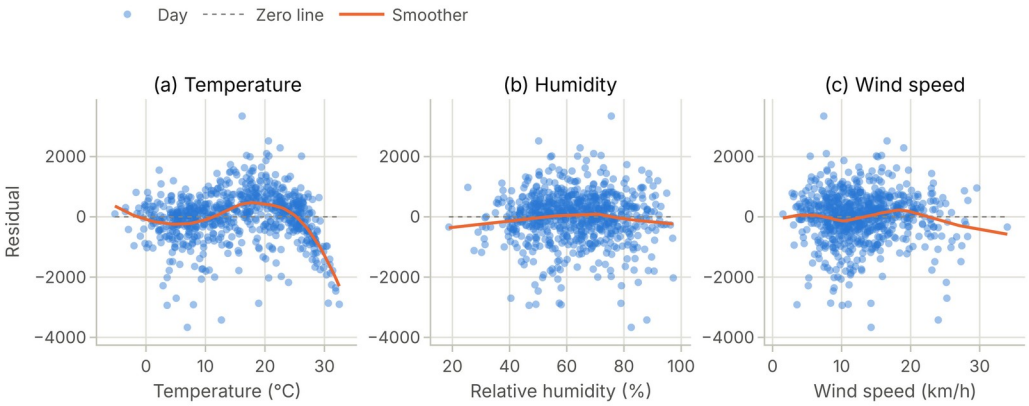


Figure 1. Residuals of the main-effects model with a linear temperature term, plotted against each weather predictor, with a smoother (orange) and the zero line. The arch for temperature (a) shows that a straight line misses the curvature; humidity (b) and wind (c) show little pattern.

A full model that adds the feels-like temperature (*atemp*) and month shows classic multicollinearity. The standard error of the temperature term more than triples (7.9 → 26.4), *atemp* is not significant ($p = 0.720$), and none of the 11 month indicators are significant. The variance inflation factors, $VIF_j = 1/(1 - R_j^2)$, are 81.1 for temperature and 72.6 for *atemp* ($r = 0.992$), and the season and month indicators reach 10.9 and 9.5 because they carry the same information (Appendix Table A2). Backward elimination by AIC drops month, then feels-like temperature, and BIC (penalty ln 730 = 6.59 per parameter) drops the same two terms (Table A3). Both therefore select

$$cnt_i = \beta_0 + \beta_1 \mathrm{Yr2012}_i + \beta_2 \mathrm{Spring}_i + \beta_3 \mathrm{Summer}_i + \beta_4 \mathrm{Fall}_i + \beta_5 \mathrm{Weekend}_i + \beta_6 \mathrm{Holiday}_i + \beta_7 \mathrm{Mist}_i + \beta_8 \mathrm{RainSnow}_i + \beta_9 T_i + \beta_{10} T_i^2 + \beta_{11} T_i^3 + \beta_{12} \mathrm{Hum}_i + \beta_{13} \mathrm{Wind}_i + \varepsilon_i, \quad \varepsilon_i \sim \text{iid } N(0, \sigma^2),$$

with each indicator compared with 2011, winter, a working day or clear weather. The model explains R^2 = 0.876 of the variation (adjusted 0.874; F = 390.1 on 13 and 716 df, $p < 0.001$), with a residual standard error of 686.1 rentals/day (Table 1).

Table 1. Final model coefficients (rentals per day), with 95 % confidence intervals estimate ± $t_{0.975,\ 716}$ × SE, where t = 1.963. *T* = temperature − 20 °C.

Term	Estimate	SE	t	p	95 % CI
(Intercept)	6,243.3	230.5	27.08	< 0.001	(5,790.8, 6,695.9)
Year 2012 (vs 2011)	1,953.6	52.03	37.55	< 0.001	(1,851.4, 2,055.7)
Spring (vs winter)	727.9	97.99	7.43	< 0.001	(535.5, 920.2)
Summer (vs winter)	985.3	126.9	7.76	< 0.001	(736.2, 1,234.4)
Fall (vs winter)	1,281.2	85.16	15.05	< 0.001	(1,114.0, 1,448.4)
Weekend (vs working day)	−95.47	56.69	−1.68	0.093	(−206.8, 15.83)
Holiday (vs working day)	−618.3	153.4	−4.03	< 0.001	(−919.4, −317.2)
Mist/cloudy (vs clear)	−371.1	69.08	−5.37	< 0.001	(−506.7, −235.5)
Light rain/snow (vs clear)	−1,741.8	183.3	−9.50	< 0.001	(−2,101.8, −1,381.9)
T (temperature − 20 °C)	72.21	7.87	9.17	< 0.001	(56.76, 87.67)
T^2	−13.72	0.790	−17.36	< 0.001	(−15.27, −12.17)
T^3	−0.481	0.040	−12.07	< 0.001	(−0.559, −0.403)
Humidity (%)	−23.76	2.65	−8.96	< 0.001	(−28.97, −18.56)
Wind speed (km/h)	−51.79	5.41	−9.57	< 0.001	(−62.41, −41.17)

3. Checking the assumptions

Figure 2 shows the four residual plots that R draws for *plot(lm)*. Standardized residuals, $e_i/(\hat{\sigma}\sqrt{(1 - h_{ii})})$, behave roughly like N(0, 1) draws if the model holds.

- **Linearity.** With the cubic term, the residuals plotted against temperature no longer arch (Appendix Figure A1), and humidity and wind are close to linear. Squared terms for humidity and wind are statistically detectable (nested F = 16.5, p < 0.001) but add only 0.005 to R^2, so the interpretable linear terms are kept. The smoother in the residuals-vs-fitted plot (Fig. 2a) is flat except at the low end. On the 26 days with fitted values below 1,000 (cold, wet or windy days) the model under-predicts by 716 rentals on average, and 3 fitted values are negative, because a linear model cannot respect the floor at zero.
- **Constant variance.** Otherwise there is no fan shape: the residual SD is 662, 663 and 698 across thirds of the fitted values, and the scale–location smoother (Fig. 2c) is nearly flat. Residuals are, however, more variable in 2012 than in 2011 (SD 782 vs 560) as the system grew, so p-values close to 0.05 deserve caution.
- **Normality.** The Q–Q plot (Fig. 2b) is straight in the middle but has a heavy left tail. 5.5 % of standardized residuals exceed ±2, close to the 5 % expected, but 11 exceed ±3, where about 2 would be expected. Nine of these are over-predictions: two hurricanes, Thanksgiving 2012 and the weekend after it, and three days coded as misty on which the hourly records show several hours of rain (Table A5). With n = 730, the t-based inference for the coefficients is still approximately valid, but single-day prediction intervals are less reliable.
- **Leverage and influence.** 53 days exceed the leverage cut-off 2p/n = 0.038 (Faraway, 2014), and 41 of them are rain or snow days or holidays, which are rare categories (20 and 21 days). The highest leverage (h = 0.133) belongs to the coldest day (−5.2 °C), whose residual is small. The largest Cook's distance is 0.112 (Hurricane Sandy, which also has the most negative residual), far below 0.5 (Fig. 2d; Appendix Table A4). Refitting without the two Sandy days moves no significant coefficient by more than 8 % (Table A6).
- **Independence.** The days are consecutive, and a residual has the same sign as the previous day's 64 % of the time (about 50 % if independent; Appendix Figure A2). Multi-day weather spells are the likely cause, so the standard errors are probably somewhat optimistic.
- **Multicollinearity.** Without *atemp* and month, all VIFs are at most 4.77 except those of the three temperature terms (7.1–12.4). T, T^2 and T^3 are functions of the same variable, so this collinearity is expected. It inflates their individual standard errors, which is why temperature is tested as a group (nested F-test) and interpreted through its curve.

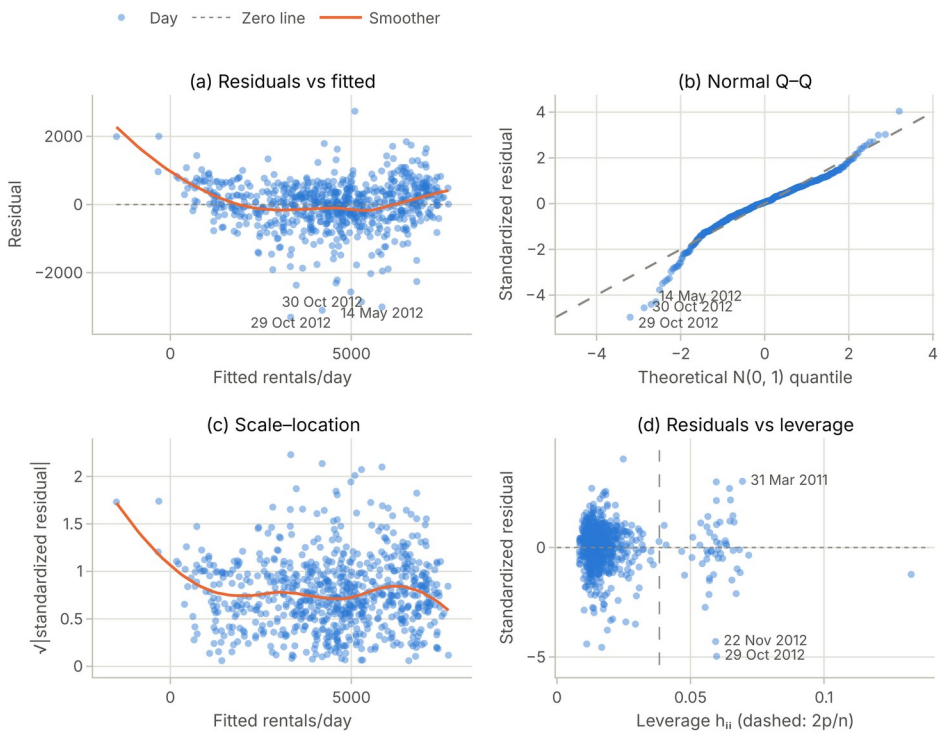


Figure 2. Residual plots for the final model, as in R's plot(lm): (a) residuals vs fitted values and (c) scale–location, with smoothers; (b) normal Q–Q plot of standardized residuals; (d) standardized residuals vs leverage (dashed line at 2p/n). The three most extreme days are labeled. No day comes close to a Cook's distance of 0.5.

4. Interpreting the key coefficients

Figure 3 shows partial effect plots (Fox & Weisberg, 2019). Each panel gives the predicted mean rentals as one predictor varies, with the numeric predictors held at their means and the factors at their observed mix. The bands are 95 % confidence intervals for the mean response, $x_0^T\hat{\beta} \pm t \cdot \hat{\sigma}\sqrt{(x_0^T(X^TX)^{-1}x_0)}$. Each statement below holds the other predictors constant.

- **Temperature** (panel a). Because temperature enters through three terms, its effect is read from the curve. An average day is predicted to have about 2,513 rentals at 0 °C, 3,986 at 10 °C and 5,599 at 20 °C, peaking at **5,687 near 22.3 °C (72 °F)**. Rentals then fall to 4,469 at 30 °C, a drop of 1,218.
- **Humidity** (−23.76 per point; panel b). Each additional percentage point of humidity means about 24 fewer rentals (95 % CI 18.6 to 29.0), or **238 fewer per 10 points**.
- **Wind** (−51.79 per km/h; panel c). A 10 km/h windier day has **518 fewer rentals** (95 % CI 412 to 624).
- **Precipitation** (panel d). Light rain or snow costs about **1,742 rentals** relative to an otherwise identical clear day (95 % CI 1,382 to 2,102). Mist or cloud costs about 371.
- **Holiday** (panel e). For H_0: $\beta_{Holiday} = 0$, $t = -618.3/153.4 = -4.03$ on 716 df ($p < 0.001$), so we reject H_0. Holidays average **618 fewer rentals than comparable working days** (95 % CI 317 to 919). Weekends do not differ significantly from working days ($t = -1.68$, $p = 0.093$).

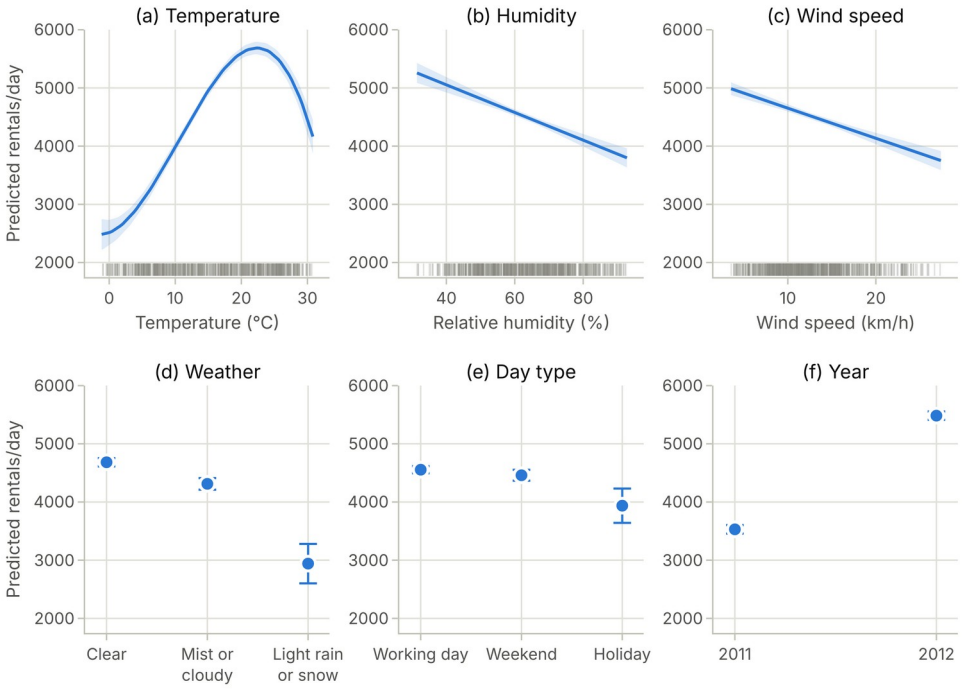


Figure 3. Partial effects from the final model with 95 % confidence bands or bars. Curves cover the central 98 % of observed values; ticks mark the observed days. Panel f shows the year control (system growth).

Prediction. For Thanksgiving 2012 (clear, 8.0 °C) the model predicts $\hat{y}_0 = 5{,}282$ rentals. With $x_0^T(X^TX)^{-1}x_0 = 0.0594$, the 95 % prediction interval for a new day like it is $5{,}282 \pm 1.963 \times 686.1\sqrt{(1 + 0.0594)} = (3{,}895,\ 6{,}668)$. The interval for the mean response is (4,953, 5,610). Only 2,425 bikes were rented (residual −2,857), while the warm Independence Day 2011 (26.2 °C) was *under*-predicted by 1,790. This suggests that the holiday effect depends on temperature, which Milestone 3 will test.

5. Conclusions and next steps

Holding season, year and the other weather variables fixed, demand peaks near 22 °C, rain or snow costs about 1,742 rentals, and holidays average 618 fewer rentals than working days. The cubic term resolves the main linearity problem, and no day is unduly influential. Several problems remain: the residuals are more variable in 2012, the left tail is heavy, residuals on consecutive days tend to share a sign, and the model under-predicts the slowest days, with negative fitted counts (minimum −1,487). These are natural targets for the count models of the final project. **Milestone 3** will use day type as the treatment in an ANCOVA with these covariates and add a holiday × temperature interaction.

References

Fanaee-T, H. (2013). *Bike Sharing* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5W894

Fanaee-T, H., & Gama, J. (2013). Event labeling combining ensemble detectors and background knowledge. *Progress in Artificial Intelligence*, 1–15. https://doi.org/10.1007/s13748-013-0040-3

Faraway, J. J. (2014). *Linear models with R* (2nd ed.). Chapman & Hall/CRC.

Fox, J., & Weisberg, S. (2019). *An R companion to applied regression* (3rd ed.). Sage.

Appendix: additional tables and diagnostics

All results are produced by the Pluto notebook *m2BaselineRegression.jl* (Julia 1.12.7), with a static HTML export *m2BaselineRegression.html* in the same folder. The notebook follows the Exam 1 review template: model statement and confidence interval (§2), coefficient and nested-model tests (§3–4), multicollinearity and VIF (§5), backward AIC/BIC (§6), and inference, prediction and residual diagnostics (§7–8).

Table A1. Model comparison. Residual SE is $\hat{\sigma} = \sqrt{(RSS/(n - p))}$.

Model	Parameters	R^2	Adj. R^2	Residual SE	AIC	BIC
Temperature only (simple regression)	2	0.394	0.393	1,505.8	12,758.6	12,772.4
Main effects, linear temperature	12	0.822	0.819	822.6	11,885.7	11,945.4
Main effects + T^2	13	0.851	0.849	752.2	11,756.1	11,820.4
Main effects + T^2 + T^3 (final)	14	0.876	0.874	686.1	11,622.8	11,691.7
Main effects + T^2 + T^3 + T^4	15	0.876	0.874	686.1	11,623.8	11,697.3
Final + feels-like temperature + month	26	0.880	0.875	682.5	11,626.8	11,750.8

Sequential F-tests: linear → quadratic $F = 141.7$ ($p < 0.001$); quadratic → cubic $F = 145.7$ ($p < 0.001$); cubic → quartic $F = 1.00$ ($p = 0.318$).

Table A2. $VIF = 1/(1 - R_i^2)$ for each coefficient of the full model (final model + feels-like temperature + month). For factors, the range over their indicator variables is shown. Values above 10 are in bold.

Term	VIF
Year 2012 (1 indicator)	1.06
Season (3 indicators)	**7.46–10.88**
Month (11 indicators)	1.99–9.46
Day type (2 indicators)	1.03–1.04
Weather (2 indicators)	1.42–1.70
T (temperature − 20 °C)	**81.10**
T^2	**16.52**
T^3	**17.23**
Feels-like temperature	**72.60**
Humidity	2.33
Wind speed	1.30

Table A3. Backward elimination path, with criteria computed as in R's step(): $n \ln(RSS/n) + k \cdot p$ with $k = 2$ (AIC) or $k = \ln n$ (BIC). At the final step no deletion lowers either criterion; removing day type would raise BIC by only 4.7.

Step	AIC	BIC
Start (full model)	9,553.17	9,672.59
Drop month	9,551.04	9,619.93
Drop feels-like temperature	9,549.19	9,613.49

Table A4. The eight days with the largest Cook's distance in the final model.

Date	Day type	Weather	Rentals	Fitted	Std. resid.	Leverage	Cook's D	What happened
Mon 29 Oct 2012	Working day	Light rain/snow	22	3,327	−4.97	0.060	0.112	Hurricane Sandy: system nearly shut down
Thu 22 Nov 2012	Holiday	Clear	2,425	5,282	−4.29	0.059	0.083	Thanksgiving
Thu 31 Mar 2011	Working day	Light rain/snow	1,685	−316	3.02	0.069	0.049	Cold, wet day; fitted value is negative
Wed 26 Jan 2011	Working day	Light rain/snow	506	−1,487	2.99	0.060	0.041	Snowstorm; fitted value is negative
Mon 4 Jul 2011	Holiday	Mist/cloudy	6,043	4,253	2.70	0.065	0.036	Independence Day
Thu 24 Nov 2011	Holiday	Clear	1,495	3,316	−2.73	0.056	0.031	Thanksgiving
Sat 17 Mar 2012	Weekend	Mist/cloudy	7,836	5,099	4.04	0.025	0.030	Warm Saturday, St. Patrick's Day
Sat 27 Aug 2011	Weekend	Mist/cloudy	1,115	3,480	−3.50	0.029	0.026	Hurricane Irene

Table A5. The days with standardized residuals beyond ±3. Rain hours counts the hourly records (*hour.csv*) with light rain or snow or worse, which the single daily weather code can hide.

Date	Day type	Weather (daily)	Rain hours	Rentals	Fitted	Std. resid.	What happened
Mon 29 Oct 2012	Working day	Light rain/snow	1	22	3,327	−4.97	Hurricane Sandy: system nearly shut down
Tue 30 Oct 2012	Working day	Mist/cloudy	6	1,096	4,197	−4.56	Hurricane Sandy, second day
Mon 14 May 2012	Working day	Mist/cloudy	8	2,843	5,849	−4.41	Coded mist, but rainy for part of the day
Thu 22 Nov 2012	Holiday	Clear	0	2,425	5,282	−4.29	Thanksgiving
Sun 25 Nov 2012	Weekend	Clear	0	2,424	4,988	−3.77	Thanksgiving weekend
Sat 27 Aug 2011	Weekend	Mist/cloudy	7	1,115	3,480	−3.50	Hurricane Irene
Sat 24 Nov 2012	Weekend	Clear	0	2,277	4,579	−3.40	Thanksgiving weekend
Sun 19 Aug 2012	Weekend	Mist/cloudy	5	4,549	6,809	−3.33	Coded mist, but rainy for part of the day
Fri 23 Sep 2011	Working day	Mist/cloudy	11	2,395	4,610	−3.26	Coded mist, but rainy for part of the day
Thu 31 Mar 2011	Working day	Light rain/snow	18	1,685	−316	3.02	Cold, wet day; fitted value is negative
Sat 17 Mar 2012	Weekend	Mist/cloudy	0	7,836	5,099	4.04	Warm Saturday, St. Patrick's Day

Table A6. Sensitivity to the two Hurricane Sandy days (29–30 October 2012): coefficients estimated with all days and without those two days.

Term	All days	SE	Without Sandy days	Change (%)
(Intercept)	6,243.3	230.5	6,116.3	−2.0
Year 2012 (vs 2011)	1,953.6	52.03	1,979.4	1.3
Spring (vs winter)	727.9	97.99	741.6	1.9
Summer (vs winter)	985.3	126.9	1,001.0	1.6
Fall (vs winter)	1,281.2	85.16	1,323.5	3.3
Weekend (vs working day)	−95.47	56.69	−108.9	−14.0
Holiday (vs working day)	−618.3	153.4	−630.8	−2.0
Mist/cloudy (vs clear)	−371.1	69.08	−372.4	−0.3
Light rain/snow (vs clear)	−1,741.8	183.3	−1,608.2	7.7
T = temperature − 20 °C	72.21	7.87	72.02	−0.3
T^2	−13.72	0.790	−13.51	1.5
T^3	−0.481	0.040	−0.474	1.5
Humidity (%)	−23.76	2.65	−22.85	3.9
Wind speed (km/h)	−51.79	5.41	−48.92	5.5

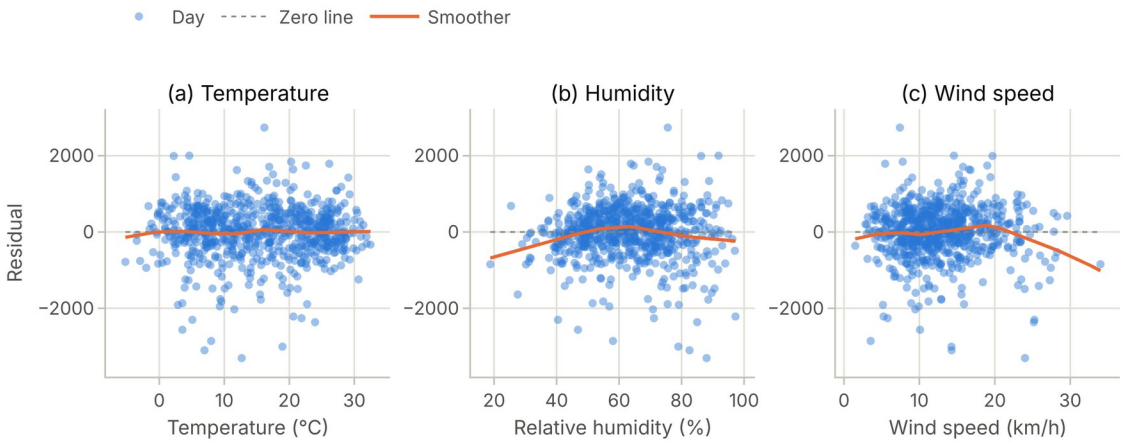


Figure A1. Residuals of the final model against each weather predictor, with a smoother (orange). The arch for temperature is gone; humidity and wind show only mild curvature at their extremes.

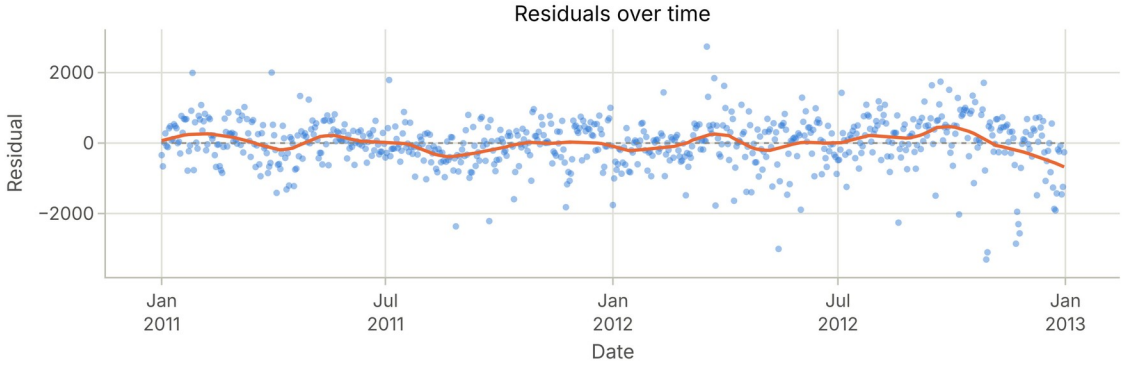


Figure A2. Residuals of the final model in time order, with a smoother. Residuals tend to stay above or below zero for several days in a row, and the large negative values in late 2012 are Hurricane Sandy and the Thanksgiving and Christmas holidays.